

Project description

The research team at my organization needs to update the file permissions for certain files and directories within the “projects” directory. The permissions do not currently reflect the level of authorization that should be given. Checking and updating these permissions will help keep their system secure and ensure the principle of least privilege is achieved. To complete this task, I performed the following tasks:

Check file and directory details

The following code demonstrates how I used Linux commands to determine the existing permissions set for a specific directory in the file system.

```
researcher2@7ea785a0a77f:~$ pwd
/home/researcher2
researcher2@7ea785a0a77f:~$ ls
projects
researcher2@7ea785a0a77f:~$ cd projects
researcher2@7ea785a0a77f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 17 11:35 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 17 11:50 ..
-rw--w---- 1 researcher2 research_team  46 Apr 17 11:35 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 17 11:35 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Apr 17 11:35 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Apr 17 11:35 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 17 11:35 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 17 11:35 project_t.txt
```

The first line displays the command I entered to list all the files and subdirectories in the projects directory, including hidden files. The other lines display the output. From the output, I discovered that there is a hidden file named project_x.txt, a subdirectory named drafts and 4 other project files. The output also displays the permissions of each file and directory, their owners and groups, the file size and the date and time of last modification. Also displayed is the block size (32) of the listed content and the number of hard links to each directory or file.

Describe the permissions string

The first 10-character string in the first column represents the permissions for each owner which can be broken down to determine who is authorized to access the file and their specific permissions.

The first character indicates the file type. It can either be (d) indicating a directory or (-) indicating a file.

The second to fourth characters indicate the read (r), write (w), and execute (x) permissions for the user respectively. When one of these characters is a hyphen (-) instead, it indicates that this permission is not granted to the user.

The fifth to seventh characters indicate the read (r), write (w), and execute (x) permissions for the group respectively. When one of these characters is a hyphen (-) instead, it indicates that this permission is not granted for the group.

The eight to tenth characters indicate the read (r), write (w), and execute (x) permissions for other respectively. This owner type consists of all other users on the system apart from the user and the group. When one of these characters is a hyphen (-) instead, that indicates that this permission is not granted for other.

Change file permissions

The organization determined that other shouldn't have write access to any of their files. To comply with this, I referred to the file permissions that I previously returned. I discovered that only the project.txt file has write access for other and must be removed in order to comply with the organizations rule.

The following code demonstrates how I used Linux commands to do this:

```
researcher2@7ea785a0a77f:~/projects$ chmod o-w project_k.txt
researcher2@7ea785a0a77f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 17 11:35 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 17 11:50 ..
-rw--w---- 1 researcher2 research_team  46 Apr 17 11:35 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 17 11:35 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Apr 17 11:35 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Apr 17 11:35 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 17 11:35 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 17 11:35 project_t.txt
```

The first two lines of the screenshot display the commands I entered, and the other lines display the output of the second command. I used the `chmod o-w project_k.txt` command to remove the write access for other in the `project_k.txt` file. After this I used the `ls -la` to list the files and confirm that the changes have been made.

Change file permissions on a hidden file

The research team at my organization recently archived `project_x.txt`. They do not want anyone to have write access to this project, but the user and group should have read access.

The following code demonstrates how I used Linux commands to change the permissions:

```
researcher2@7ea785a0a77f:~/projects$ chmod u=r,g=r .project_x.txt
researcher2@7ea785a0a77f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 17 11:35 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 17 11:50 ..
-r--r----- 1 researcher2 research_team  46 Apr 17 11:35 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 17 11:35 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Apr 17 11:35 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Apr 17 11:35 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 17 11:35 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 17 11:35 project_t.txt
```

The first two lines of the screenshot display the commands I entered, and the other lines display the output of the second command. I know `.project_x.txt` is a hidden file because it starts with a period (`.`). In this example, since the goal is to grant only read access for both user and groups, I simply used the `chmod u=r,g=r .project_x.txt` which overwrites all existing permissions and replaces it with the one specified (`r`).

Change directory permissions

My organization only wants the `researcher2` user to have access to the `drafts` directory and its contents. This means that no one other than `researcher2` should have execute permissions.

The following code demonstrates how I used Linux commands to change the permissions:

```
researcher2@721ab73f75e3:~/projects$ chmod g-x drafts
researcher2@721ab73f75e3:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 17 11:48 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 17 12:59 ..
-rw--w---- 1 researcher2 research_team  46 Apr 17 11:48 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Apr 17 11:48 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Apr 17 11:48 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Apr 17 11:48 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 17 11:48 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 17 11:48 project_t.txt
researcher2@721ab73f75e3:~/projects$
```

The first two lines of the screenshot display the commands I entered, and the other lines display the output of the second command. Previously the group had execute rights for the `drafts` subdirectory by running the command `chmod g-x drafts`, I was able to remove the execute rights from it, leaving only the user/owner with full access privileges.

Summary

I changed multiple permissions to match the level of authorization my organization wanted for files and directories in the `projects` directory. The first step in this was using `ls -la` to check the permissions for the directory. This informed my decisions in the following steps. I then used the

chmod command with the required arguments to change the permissions on files and directories.